

Adaptive Experimentation: Alibaba and Amazon

Week 3

The Dynamic Pricing Problem

A retailer that wants to maximize revenue needs to know how demand for each product varies with price. For a store with a handful of products, a manager can test a few price points and develop reasonable intuition. But a large e-commerce platform may list millions of SKUs, and the relationship between price and demand for each one depends on inventory levels, competitor prices, seasonality, and customer preferences. No team of analysts can estimate these relationships across millions of products, and any fixed rule for setting prices will likely ignore much of the relevant information.

Rules-based approaches, such as matching a competitor's listed price, or adding a fixed markup to cost, are simple to implement but ignore the information contained in customer responses. The harder problem is learning which prices "work" -- testing them, observing how customers respond, and adjusting accordingly. A/B testing is the standard tool for this kind of learning, but as Amazon and Alibaba both discovered, pricing creates complications that standard tests struggle to handle.

Learning One Price at a Time

In 2013, Amazon was reported to change prices on its platform roughly 2.5 million times per day, about 46 times more often than Walmart. Each change is also an observation: it generates data on how customers respond at that price, and retailers that experiment with prices frequently stand to learn more about the market. This volume of price changes provides Amazon with an informational advantage, on top of their operational ones.

Alibaba faced the same problem on the other side of the Pacific. In 2018, the company's Supply Chain Platform team deployed an automated pricing system on Tmall.com, China's largest business-to-consumer platform. The system covered two distinct settings. The first was markdown pricing for limited-inventory luxury goods, mainly handbags and clothes, with roughly 500 SKUs with about 10 items each in stock. The second was daily pricing for fast-moving consumer goods, about 1,000 SKUs drawn from 200 product categories with effectively unlimited supply. For each product, the model tracked four groups of signals: price variables (discount rate, coupons, actual payment), sales volume and revenue, customer traffic (unique visitors, page views, buyer counts), and competitiveness indicators (reviews, similar products' prices). It set a new price once per day.

A critical design choice was how to measure whether a price was “working.” Total revenue was misleading because daily customer traffic fluctuated so much that it swamped any signal from price changes. The team instead used revenue per unique visitor, which isolated the price effect from traffic swings.

In the daily-pricing experiment, the algorithm’s revenue-per-visitor growth was roughly five times that of the manually-priced control group. In the markdown experiment on luxury products, the algorithm kept profit margins positive throughout most of a 15-day clearance period, while human pricers drove margins negative by discounting too aggressively. It was the first reported deployment of **reinforcement learning** for real-time pricing on a major e-commerce platform.

Discussion Questions

Question 1. Suppose Alibaba ran a traditional A/B test, splitting traffic evenly across a handful of candidate prices for a fixed window and then picking the winner. What are some of the disadvantages of this approach?

Question 2. Suppose Alibaba’s system currently believes that \$25 is the best price for a particular product. What factors should determine how far from the current best price the system is willing to experiment?

Question 3. Should Alibaba experiment more aggressively on an established household staple or on a newly launched product? Why?

Sources

Liu, Jiayi, et al. “Dynamic Pricing on E-commerce Platform with Deep Reinforcement Learning: A Field Experiment.” arXiv:1912.02572, December 2019. arxiv.org/abs/1912.02572

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